

```
System.out.println("fib(20) iterative = " + iterative  
(20) + "name would need ~10^19 calls");  
}
```

Fibonacci using recursion:-

→ $n=5$ $n < 2 \Rightarrow \text{false}$ $\text{name}(0) = 0$ $\text{name}(1) = 1$
 $\text{name}(5) = \text{name}(4) + \text{name}(3) = 2 + 3 = 5$
 $\text{name}(4) = \text{name}(3) + \text{name}(2) = 2 + 1 = 3$
 $\text{name}(3) = \text{name}(2) + \text{name}(1) = 1 + 1 = 2$
 $\text{name}(2) = \text{name}(1) + \text{name}(0) = 1 + 0 = 1$
fibonacci sequence = 0, 1, 1, 2, 3, 5

Fibonacci using loops:-

→ $a=0$ $b=1$ $t=a+b=0+1=1$ sequence: 0 1 1
 $a=b=1$ $b=t=1$ $t=a+b=1+1=2$ 0 1 1 2
 $a=b=1$ $b=t=2$ $t=a+b=1+2=3$ 0 1 1 2 3
 $a=b=2$ $b=t=3$ $t=a+b=2+3=5$ 0 1 1 2 3 5

Galaxy M13

Recursion:-



printSeries:-

→ printSeries(10) to printSeries(1) ⇒ no output
printSeries(11) → printSeries(10) → 10
printSeries(9) → 9
printSeries(8) → 8
printSeries(7) → 7
printSeries(6) → 6

printSeries (5) $\rightarrow$ 5
printSeries (4) $\rightarrow$ 4
printSeries (3) $\rightarrow$ 3
printSeries (2) $\rightarrow$ 2
printSeries (1) $\rightarrow$ 1

diffOfTime :-

$\rightarrow$ diffOfTime (0) $\rightarrow$ 0
diffOfTime (2) $\rightarrow$ $0+2=2$
diffOfTime (4) $\rightarrow$ ~~$0+4=4$~~ $2+2=4$
diffOfTime (6) $\rightarrow$ $4+2=6$
diffOfTime (8) $\rightarrow$ $6+2=8$
diffOfTime (10) $\rightarrow$ $8+2=10$
diffOfTime (12) $\rightarrow$ $10+2=12$
diffOfTime (14) $\rightarrow$ $12+2=14$
diffOfTime (16) $\rightarrow$ $14+2=16$
diffOfTime (18) $\rightarrow$ $16+2=18$
diffOfTime (20) $\rightarrow$ $18+2=20$

tables :-

$\rightarrow$ tables (2,1) $\rightarrow$ $2 \times 1 = 2$
tables (2,2) $\rightarrow$ $2 \times 2 = 4$
tables (2,3) $\rightarrow$ $2 \times 3 = 6$
tables (2,4) $\rightarrow$ $2 \times 4 = 8$
tables (2,5) $\rightarrow$ $2 \times 5 = 10$
tables (2,6) $\rightarrow$ $2 \times 6 = 12$
tables (2,7) $\rightarrow$ $2 \times 7 = 14$
tables (2,8) $\rightarrow$ $2 \times 8 = 16$
tables (2,9) $\rightarrow$ $2 \times 9 = 18$
tables (2,10) $\rightarrow$ $2 \times 10 = 20$